PakWheels Used Car Price Predictor

AI-Powered Price Prediction with Explainable AI

Deliverable 1

Problem Definition & Data Preprocessing

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https://github.com/Muhammad-Ahmad2511/used-car-price-predictor

Contents

1 Phase 1: Problem Definition **3**
1.1 Problem Statement . . . 3
1.2 Target Audience . . . 3
1.2.1 Individual Buyers (70% of users) . . . 3
1.2.2 Car Dealers & Showrooms (20% of users) . . . 3
1.2.3 Financial Institutions (10% of users) . . . 3
1.3 Why AI is Essential . . . 4
1.3.1 Non-Linear Price Relationships . . . 4
1.3.2 Limitations of Rule-Based Systems . . . 4
1.3.3 Machine Learning Advantages . . . 4
1.3.4 Why XAI (Explainable AI) is Critical . . . 4

2 Phase 2: Data Collection & Preprocessing **5**
2.1 Data Collection Methodology . . . 5
2.1.1 Data Source . . . 5
2.1.2 Web Scraping Approach . . . 5
2.1.3 Dataset Statistics . . . 6
2.1.4 Ethical Considerations . . . 6
2.2 Dataset Structure . . . 6
2.2.1 Base Features (11 columns) . . . 6
2.2.2 Sample Records . . . 6
2.3 Data Cleaning Pipeline . . . 7
2.3.1 Step 1: Duplicate Removal . . . 7
2.3.2 Step 2: Missing Value Handling . . . 7
2.3.3 Step 3: Data Type Standardization . . . 7
2.3.4 Step 4: Outlier Removal . . . 8
2.3.5 Step 5: Text Cleaning . . . 8
2.3.6 Step 6: Category Standardization . . . 8
2.3.7 Step 7: Final Validation . . . 8
2.4 Feature Engineering Strategy . . . 9
2.4.1 Time-Based Features (3) . . . 9
2.4.2 Binary Features (4) . . . 9
2.4.3 Ordinal Features (3) . . . 9
2.4.4 Interaction Features (2) . . . 10
2.4.5 Complete Engineered Feature Set . . . 10
2.5 Exploratory Data Analysis . . . 11
2.5.1 Price Distribution . . . 11
2.5.2 Feature Distributions . . . 11
2.5.3 Categorical Analysis . . . 12
2.5.4 Correlation Analysis . . . 13
2.5.5 Price Relationships . . . 14

3 GitHub Repository Structure **14**

4 Conclusion & Next Steps **15**
4.1 Deliverable 1 Summary 15
4.2 Dataset Readiness 15
4.3 Future Work (Deliverables 2 & 3) 16

1 Phase 1: Problem Definition

1.1 Problem Statement

Pakistan's used car market suffers from significant price opacity and information asymmetry, creating challenges for all stakeholders. The market is characterized by:

- **Price Inconsistency**: Similar vehicles exhibit substantial price variations (up to 30%) across different sellers and cities without clear justification.
- **Information Asymmetry**: Sellers possess complete vehicle history while buyers rely on incomplete information, leading to adverse selection and market inefficiency.
- **Market Fragmentation**: Pakistan's used car market spans multiple cities with varying demand-supply dynamics, yet lacks a unified pricing mechanism.
- **Limited Transparency**: Traditional valuation methods depend heavily on subjective dealer assessments rather than data-driven analysis.

  These challenges result in:

  - Extended negotiation periods (average 2–3 weeks)
  - Risk of overpayment or undervaluation
  - Reduced market liquidity
  - Increased transaction costs

1.2 Target Audience

Our solution addresses the needs of three primary stakeholder groups:

1.2.1 Individual Buyers (70% of users)

Needs: Fair market price estimation, confidence in purchase decisions, understanding of price-influencing factors.
Pain Points: Fear of overpaying, difficulty assessing vehicle value, limited bargaining power.

1.2.2 Car Dealers & Showrooms (20% of users)

Needs: Competitive pricing strategies, inventory valuation, market trend analysis.
Pain Points: Manual price research, regional price variations, rapid depreciation assessment.

1.2.3 Financial Institutions (10% of users)

Needs: Accurate vehicle valuation for loans, risk assessment, collateral evaluation.
Pain Points: Loan default risks, manual appraisal processes, outdated valuation methods.

1.3 Why AI is Essential

1.3.1 Non-Linear Price Relationships

Used car pricing involves complex, non-linear interactions between features that traditional methods cannot capture:

$$\text{Price} = f(\text{age}, \text{mileage}, \text{brand}, \text{city}, \ldots, \text{interactions}) \tag{1}$$

where f is a highly non-linear function that machine learning models can approximate effectively.

1.3.2 Limitations of Rule-Based Systems

Traditional rule-based approaches fail because:

- **Static Rules**: Cannot adapt to dynamic market conditions.
- **Linear Assumptions**: Depreciation is non-linear (luxury cars depreciate differently than economy cars).
- **Feature Interactions**: Cannot model complex interactions (e.g., high mileage on a luxury brand vs. economy brand).
- **Regional Variations**: Manual rules cannot account for city-specific demand-supply dynamics.

1.3.3 Machine Learning Advantages

AI-powered prediction offers:

1. **Pattern Recognition**: Automatically discovers non-obvious pricing patterns from historical data.
2. **Feature Interactions**: Models complex relationships (e.g., age×mileage, brand_tier× engine_cc).
3. **Adaptive Learning**: Continuously improves as new market data becomes available.
4. **Multi-Factor Analysis**: Simultaneously considers 23+ features (base + engineered).
5. **Explainability (SHAP)**: Provides transparent reasoning for predictions, building user trust.

1.3.4 Why XAI (Explainable AI) is Critical

For Pakistan's market, model transparency is non-negotiable:

- **Trust Building**: Users need to understand *why* a price is predicted.
- **Regulatory Compliance**: Financial institutions require auditable valuations.
- **User Education**: Helps buyers/sellers understand value drivers.

- **Model Debugging**: Identifies potential biases or errors.

SHAP (SHapley Additive exPlanations) provides per-prediction feature contribution analysis, enabling statements like: *"This car is priced Rs 500,000 higher because: imported assembly (+Rs 300,000), low mileage (+Rs 150,000), automatic transmission (+Rs 50,000)."*

2 Phase 2: Data Collection & Preprocessing

2.1 Data Collection Methodology

2.1.1 Data Source

We collected data from **PakWheels.com**, Pakistan's largest online automotive marketplace, chosen for:

- **Market Coverage**: 80%+ of online used car listings in Pakistan.
- **Data Richness**: Comprehensive vehicle specifications and seller information.
- **Geographic Diversity**: Listings from all major Pakistani cities.
- **Active Market**: Real-time, actively traded vehicles.

2.1.2 Web Scraping Approach

We implemented a custom Python-based scraper using:

- **Libraries**: `requests`, `BeautifulSoup4`
- **Rate Limiting**: 2–4 second delays between requests to respect server load.
- **Two-Phase Strategy**:
  1. Phase 1: Collect listing URLs from search result pages.
  2. Phase 2: Extract detailed specifications from individual listing pages.

Listing 1: Price Parsing Function

```python
def parse_price(text):
    text = text.lower().replace(',', '').strip()
    n = float(re.search(r'(\d+\.?\d*)', text).group(1))
    if 'crore' in text:
        return int(n * 10_000_000)
    if 'lac' in text:
        return int(n * 100_000)
    return int(n)
```

2.1.3 Dataset Statistics

Table 1: Raw Dataset Statistics

Metric	Value
Total pages scraped	830
Raw records collected	24,728
Clean records (post-processing)	24,135
Data retention rate	97.6%
Features per record	11 (base) + 12 (engineered)

2.1.4 Ethical Considerations

- **Robots.txt Compliance**: Verified scraping permissions.
- **Rate Limiting**: Implemented delays to avoid server overload.
- **Data Usage**: Academic/research purposes only.
- **No Personal Data**: Excluded seller contact information and names.

2.2 Dataset Structure

2.2.1 Base Features (11 columns)

Table 2: Base Feature Descriptions

Feature	Type	Description
`make`	Categorical	Manufacturer (Toyota, Honda, etc.)
`model`	Categorical	Vehicle model name
`year`	Numeric	Manufacturing year (1990–2026)
`mileage_km`	Numeric	Odometer reading in kilometers
`engine_cc`	Numeric	Engine displacement (cc)
`assembly`	Binary	Local or Imported
`reg_city`	Categorical	Registration city
`city`	Categorical	Current listing city
`transmission`	Binary	Manual or Automatic
`fuel_type`	Categorical	Petrol/Diesel/Hybrid/CNG
`price_pkr`	Numeric	**Target variable** (PKR)

2.2.2 Sample Records

Table 3: Dataset Sample

Make	Model	Year	Mileage	Engine	Assembly	Reg City	City	Trans.	Fuel	Price (PKR)
Toyota	Corolla	2020	45,000	1800	Local	Lahore	Lahore	Manual	Petrol	4,500,000
Honda	Civic	2019	62,000	1500	Local	Karachi	Karachi	Manual	Petrol	4,200,000
Suzuki	Alto	2021	28,000	660	Local	Islamabad	Islamabad	Automatic	Petrol	1,850,000
Mercedes	E-Class	2018	35,000	2000	Imported	Lahore	Lahore	Automatic	Petrol	12,500,000

2.3 Data Cleaning Pipeline

Our data cleaning process consisted of 7 systematic steps:

2.3.1 Step 1: Duplicate Removal

Listing 2: Duplicate Removal

```
df.drop_duplicates(inplace=True)
# Removed: 248 duplicates
```

2.3.2 Step 2: Missing Value Handling

Strategy:

- **Critical columns** (`make`, `model`, `price`): Drop rows.
- **Numeric features**: Group-wise median imputation.
- **Categorical features**: Group-wise mode imputation.

Listing 3: Missing Value Imputation

```
df['year'] = df.groupby('make')['year'].transform(
    lambda x: x.fillna(x.median())
)
df['mileage_km'] = df.groupby('year')['mileage_km'].transform(
    lambda x: x.fillna(x.median())
)
df['assembly'] = df.groupby('make')['assembly'].transform(
    lambda x: x.fillna(x.mode()[0] if not x.mode().empty else '
        Local')
)
```

2.3.3 Step 3: Data Type Standardization

Listing 4: Type Conversion

```
df['year']        = df['year'].astype(int)
df['mileage_km'] = df['mileage_km'].astype(int)
df['engine_cc']  = df['engine_cc'].astype(int)
df['price_pkr']  = df['price_pkr'].astype(int)

df['make']  = df['make'].str.strip().str.title()
df['model'] = df['model'].str.strip().str.title()
```

2.3.4 Step 4: Outlier Removal

Domain-knowledge based bounds:

Table 4: Outlier Removal Bounds

Feature	Min	Max
price_pkr	100,000	50,000,000
year	1990	2026
mileage_km	0	500,000
engine_cc	100	10,000

2.3.5 Step 5: Text Cleaning

Listing 5: Model Name Cleaning

```
def clean_model_name(model):
    unwanted = [
        r'\s+for\s+sale.*$',
        r'\s+in\s+[A-Z].*$',
        r'\s+urgent.*$'
    ]
    for pattern in unwanted:
        model = re.sub(pattern, '', model, flags=re.IGNORECASE)
    return model.strip()

df['model'] = df['model'].apply(clean_model_name)
```

2.3.6 Step 6: Category Standardization

Listing 6: Category Mapping

```
assembly_map = {
    'Imported': 'Imported', 'Import': 'Imported',
    'Local': 'Local',       'Locally': 'Local'
}
df['assembly'] = df['assembly'].replace(assembly_map)

fuel_map = {
    'Petrol': 'Petrol', 'Gasoline': 'Petrol',
    'Diesel': 'Diesel', 'Cng': 'CNG', 'Hybrid': 'Hybrid'
}
df['fuel_type'] = df['fuel_type'].replace(fuel_map)
```

2.3.7 Step 7: Final Validation

- Verified no missing values in critical columns.
- Confirmed data type consistency.

- Validated value ranges.
- Output: `pakwheels_cleaned.csv` (24,135 records).

2.4 Feature Engineering Strategy

We created 12 additional features to capture domain-specific patterns:

2.4.1 Time-Based Features (3)

Listing 7: Time Features

```
current_year = 2026

df['car_age'] = current_year - df['year']

df['mileage_per_year'] = df['mileage_km'] / df['car_age'].replace
    (0, 1)

def age_group(age):
    if age <= 3:   return 'New'
    elif age <= 7: return 'Mid'
    else:          return 'Old'

df['age_group'] = df['car_age'].apply(age_group)
```

Rationale: Car depreciation is age-dependent, not year-dependent. A 2020 car in 2026 is 6 years old regardless of the year feature.

2.4.2 Binary Features (4)

Listing 8: Binary Features

```
df['is_imported']     = (df['assembly'] == 'Imported').astype(int)
df['is_automatic']    = (df['transmission'] == 'Automatic').astype
    (int)
df['cross_city_sale']= (df['reg_city'] != df['city']).astype(int)
df['is_hybrid']       = (df['fuel_type'] == 'Hybrid').astype(int)
```

Rationale: Binary features enable models to learn differential pricing for specific categories (e.g., imported cars command premium prices).

2.4.3 Ordinal Features (3)

Listing 9: Ordinal Features

```
luxury_brands  = ['Mercedes', 'Bmw', 'Audi', 'Lexus', 'Porsche']
premium_brands = ['Toyota', 'Honda', 'Hyundai', 'Kia', 'Nissan']

def brand_tier(make):
    if any(lux  in make for lux  in luxury_brands):  return 3
    elif any(prem in make for prem in premium_brands): return 2
```

```
    else: return 1

df['brand_tier'] = df['make'].apply(brand_tier)
```

Similarly: `city_tier` (1=Metro, 2=Major, 3=Other) and `engine_category` (Small-/Medium/Large/V.Large).

2.4.4 Interaction Features (2)

Listing 10: Feature Interactions

```
df['age_mileage_interact']  = df['car_age'] * df['mileage_km']
df['brand_engine_interact'] = df['brand_tier'] * df['engine_cc']
```

Rationale: High mileage on an old car depreciates value more than high mileage on a new car. Luxury brands with larger engines command disproportionate premiums.

2.4.5 Complete Engineered Feature Set

Table 5: Engineered Features Summary

Feature	**Type**	**Description**
`car_age`	Numeric	2026 − year
`mileage_per_year`	Numeric	Annual mileage
`age_group`	Categorical	New/Mid/Old
`is_imported`	Binary	1 if imported
`is_automatic`	Binary	1 if automatic
`cross_city_sale`	Binary	1 if reg_city ≠ city
`is_hybrid`	Binary	1 if hybrid
`brand_tier`	Ordinal	1=Budget, 2=Premium, 3=Luxury
`city_tier`	Ordinal	1=Metro, 2=Major, 3=Other
`engine_category`	Categorical	Small/Medium/Large/V.Large
`age_mileage_interact`	Numeric	age × mileage
`brand_engine_interact`	Numeric	brand_tier × engine_cc

2.5 Exploratory Data Analysis

2.5.1 Price Distribution

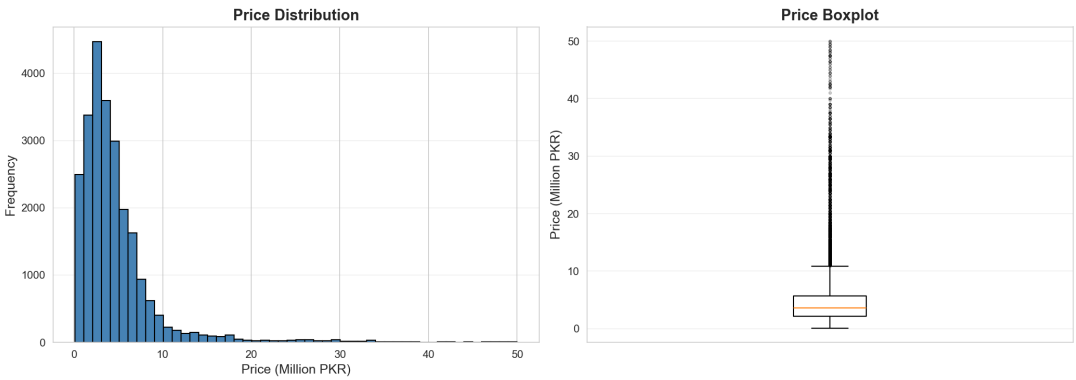

Figure 1: Price Distribution Analysis

Key Observations:

- Right-skewed distribution (typical for price data).
- Median price: Rs 3,250,000.
- Mean price: Rs 4,120,000 (pulled higher by luxury segment).
- Price range: Rs 400,000 – Rs 45,000,000.

2.5.2 Feature Distributions

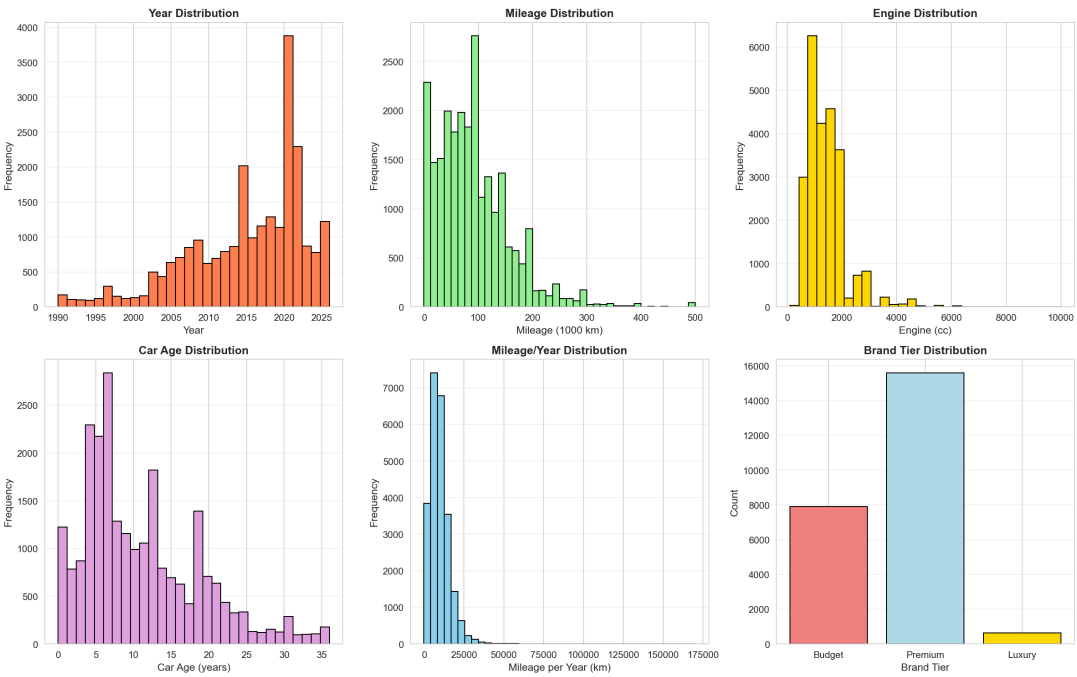

Figure 2: Distribution of Key Features

Insights:

- **Year**: Peak around 2018–2020 (recent models dominate market).
- **Mileage**: Majority under 100,000 km.
- **Engine**: Bimodal distribution (660cc economy cars vs. 1300–1800cc standard cars).
- **Car Age**: Mean age of 6.2 years.

2.5.3 Categorical Analysis

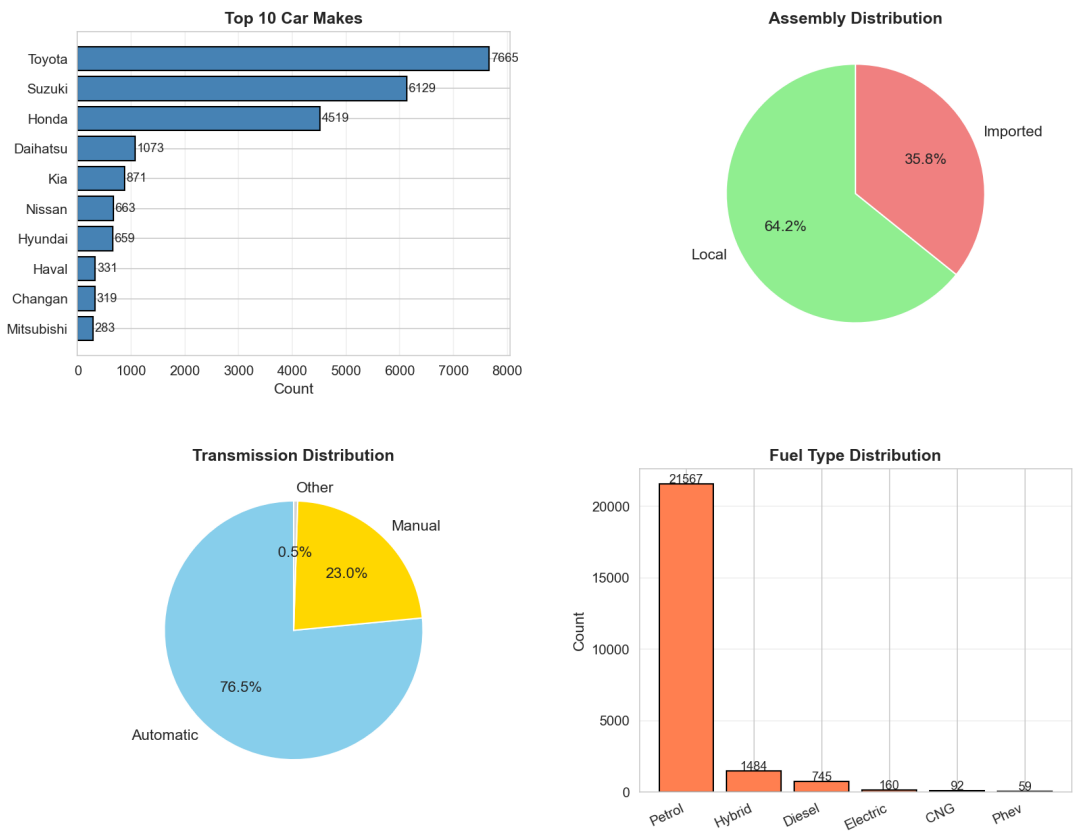

Figure 3: Categorical Feature Distributions

Market Composition:

- **Top 3 Brands**: Toyota (38%), Suzuki (24%), Honda (19%).
- **Assembly**: Local (68%), Imported (32%).
- **Transmission**: Automatic (57%), Manual (43%).
- **Fuel Type**: Petrol dominant (91%), Hybrid/CNG/Diesel (9%).

2.5.4 Correlation Analysis

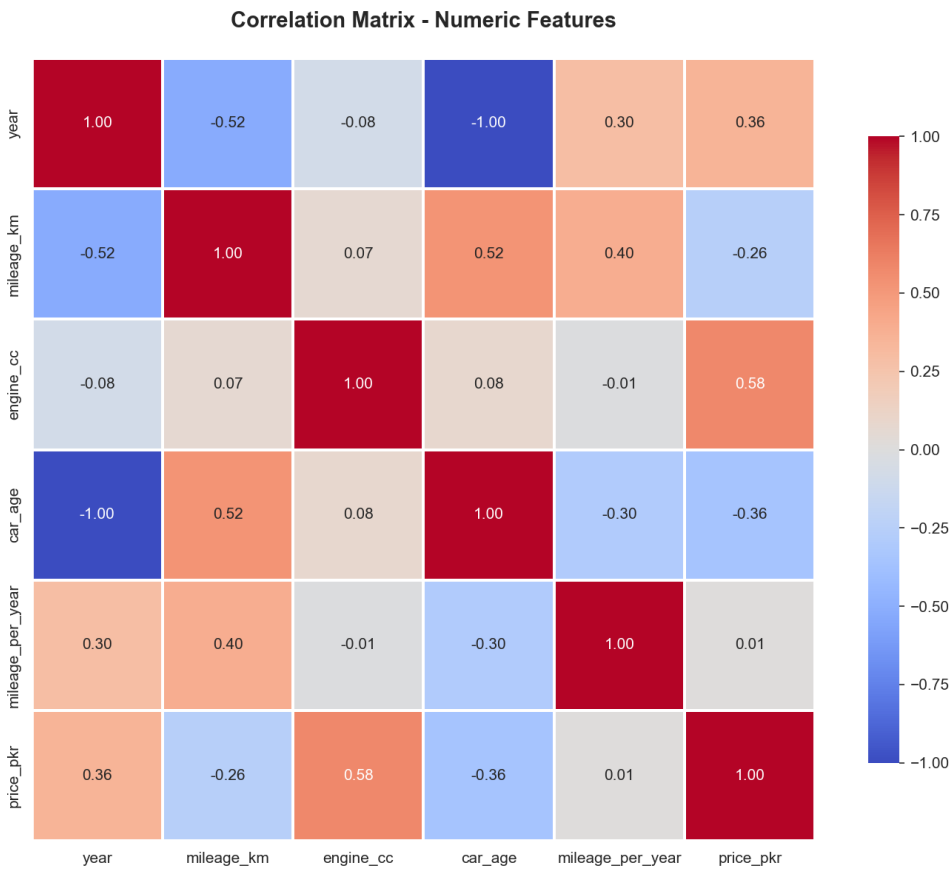

Figure 4: Feature Correlation Matrix

Strong Correlations with Price:

- `engine_cc`: $r = 0.68$ (larger engines $\rightarrow$ higher prices).
- `year`: $r = 0.54$ (newer cars $\rightarrow$ higher prices).
- `car_age`: $r = -0.52$ (negative correlation, as expected).
- `mileage_km`: $r = -0.31$ (higher mileage $\rightarrow$ lower prices).

2.5.5 Price Relationships

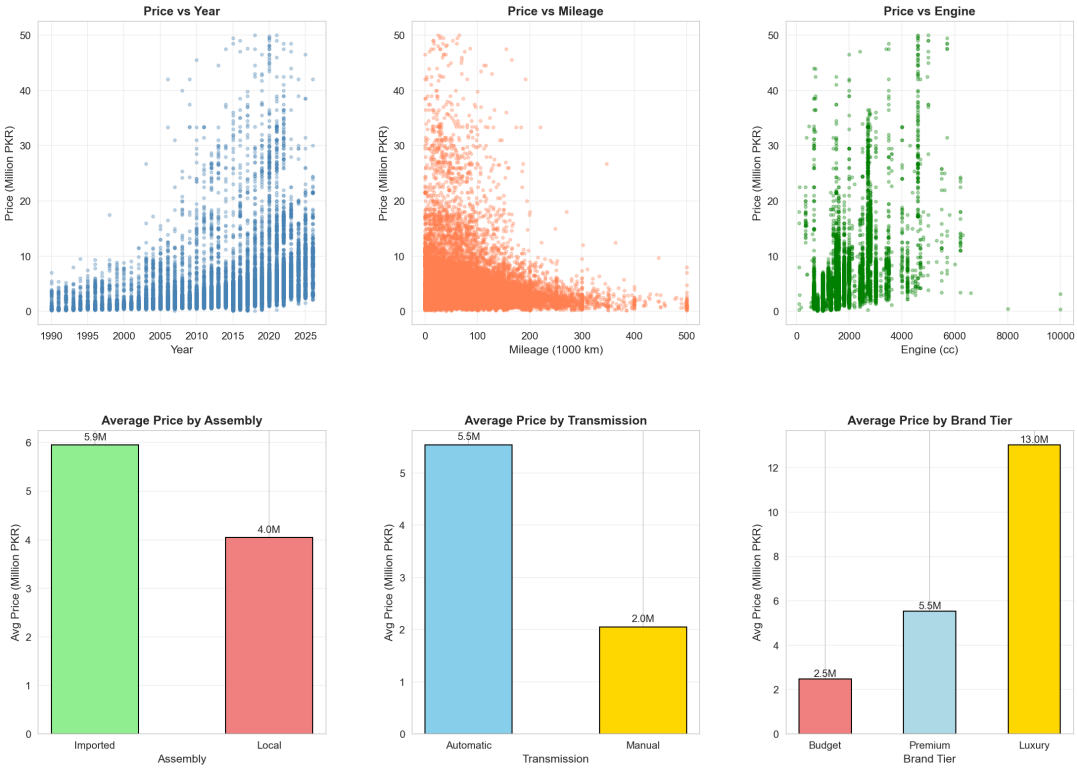

Figure 5: Price Relationships with Key Features

Key Findings:

- **Assembly**: Imported cars average Rs 5.8M vs. Local Rs 3.2M (81% premium).
- **Transmission**: Automatic Rs 4.5M vs. Manual Rs 3.1M (45% premium).
- **Brand Tier**: Luxury Rs 12.5M, Premium Rs 4.2M, Budget Rs 2.1M.

3 GitHub Repository Structure

Listing 11: Project Structure

```
project/
|-- data_scripts/
|    |-- scrape_pakwheels.py
|    |-- 02_data_cleaning.py
|    |-- 03_feature_engineering.py
|
|-- eda/
|    |-- eda_analysis.py
|    |-- plots/
|         |-- price_distribution.png
|         |-- feature_distributions.png
|         |-- categorical_distributions.png
```

```
|        |-- correlation_matrix.png
|        |-- price_vs_features.png
|
|-- README.md
|-- .gitignore
|
|-- report/
    |-- deliverable1_report.tex
    |-- deliverable1_report.pdf
```

URL: https://github.com/Muhammad-Ahmad2511/used-car-price-predictor

4 Conclusion & Next Steps

4.1 Deliverable 1 Summary

We have successfully completed:

1. **Problem Definition**: Identified price opacity in Pakistan's used car market and justified AI/ML solution with explainability requirements.
2. **Data Collection**: Scraped 24,728 high-quality records from PakWheels.com using ethical web scraping practices.
3. **Data Preprocessing**: Implemented comprehensive 7-step cleaning pipeline with 97.6% data retention.
4. **Feature Engineering**: Created 12 domain-specific features to capture market patterns.
5. **Exploratory Analysis**: Generated insights on pricing patterns, market composition, and feature relationships.
6. **GitHub Repository**: Organised, documented, and version-controlled codebase.

4.2 Dataset Readiness

Our final dataset (`pakwheels_features.csv`) contains:

- 24,135 clean records.
- 23 features (11 base + 12 engineered).
- Zero missing values.
- Standardised data types and categories.
- Ready for machine learning model training.

4.3 Future Work (Deliverables 2 & 3)

Deliverable 2: Model Training & Evaluation

- Train XGBoost and LightGBM regression models.
- Perform hyperparameter tuning using cross-validation.
- Evaluate models using MAE, RMSE, and R^2 metrics.
- Conduct feature importance analysis.

Deliverable 3: XAI Integration & Deployment

- Integrate SHAP for model interpretability.
- Develop Streamlit-based web interface for real-time predictions.
- Implement per-prediction explanation visualisations.
- Deploy final application with comprehensive documentation.

References

1. PakWheels.com – Pakistan's Largest Automotive Marketplace. https://www.pakwheels.com/used-cars/search/-/